Andres Nava

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Physics PhD student researching theoretical machine learning and the physics of learning, with experience in representation learning, scientific deep learning, and large-scale ML pipelines.

Education

Johns Hopkins University — Baltimore, MD
Ph.D. in Physics, Department of Physics & Astronomy — *Aug. 2025 - Present*
Expected completion: June 2029

California Institute of Technology — Pasadena, CA
B.S. in Physics; GPA: 3.8 — *Sep. 2021 - Mar. 2025*

Research Experience

Johns Hopkins University — Baltimore, MD
Graduate Researcher, Simons Collaboration on the Physics of Learning — *Aug. 2025 - Present*

- Investigating the physics of learning: how data structure, model architecture, and learning dynamics shape representations and computation in neural systems.
- Developed a spectral theory linking word co-occurrence statistics to coarse-to-fine hierarchical concept geometry; tested predictions in word2vec embeddings and LLM representations.
- First author of a **NeurIPS 2026** paper on concept geometry, also presented at the ICML 2026 Mechanistic Interpretability Workshop; co-author of an **ICML 2026 Spotlight**.

California Institute of Technology, LIGO Laboratory — Pasadena, CA
Undergraduate Researcher — *Jan. 2024 - Mar. 2025*

- Developed representation-learning methods in **PyTorch and JAX** to identify parameter degeneracies in binary black hole gravitational-wave signals.
- Combined learned embeddings with **symbolic regression (PySR)** to derive interpretable coordinate transformations that decouple nonlinear correlations in astrophysical parameter space.

California Institute of Technology, CMS Group — Pasadena, CA
Undergraduate Researcher — *Dec. 2021 - May 2024*

- Built end-to-end ML pipelines with Coffea and Scikit-HEP for Higgs boson pair analysis on **petabyte-scale CMS Run 2 data**, spanning preprocessing, feature engineering, modeling, and evaluation.
- Benchmarked **graph neural networks and transformers** for boosted jet classification, contributing to CMS analyses HIG-23-012 and B2G-23-007.

Publications & Presentations

Hierarchical Concept Geometry in Language Models Emerges from Word Co-occurrence
Andres Nava, Matthieu Wyart. arXiv:2605.23821 | Code
NeurIPS 2026; also presented at the ICML 2026 Mechanistic Interpretability Workshop (poster).

Symmetry in Language Statistics Shapes the Geometry of Model Representations
Dhruva Karkada, Daniel J. Korchinski, **Andres Nava**, Matthieu Wyart, Yasaman Bahri. arXiv:2602.15029
ICML 2026 Spotlight.

Using Symbolic Regression to Characterize Degeneracies in Binary Black Hole Parameter Space
Andres Nava, Katerina Chatziioannou, Aaron Johnson, Simona Miller, Lucy Thomas. ADS abstract
Oral presentation, 245th Meeting of the American Astronomical Society, Jan. 2025.

Technical Skills

ML / Scientific Computing: PyTorch, JAX, scikit-learn, NumPy, Pandas, PySR
Languages: Python, Julia, MATLAB, Mathematica, Java
Tools: Git, Docker, Coffea, Scikit-HEP, Matplotlib, LaTeX

Awards

Caltech LIGO Undergraduate Research Fellow; Haren Lee Fisher Memorial Award in Junior Physics; Carl F. Braun SURF Fellow (2x); Hispanic Scholarship Fund Scholar (3x).